Sun
30/10/2022

UNIT - II

Basic Transformations and Image Interpolation

Topics:

Translation
Rotation
Scaling
} 2-D Transformations (or) 2-D Geometric Transformations

Camera Model and Image Geometry
Camera Calibration and Stereo Imaging
Stereo Image Modeling
Interpolation and Resampling
B-spline interpolation functions
Constant Interpolation
Image Transformation
DCT Basis Images
Walsh Transform
Hadamard Transform

2-D Transformation (or) 2-D Geometric Transformations

* Various transformation functions can be applied to manipulate an object in 2-D space.

* These transformations can change the position, size and orientation of an object.

* The basic 2-D transformations are
  - Translation
  - Scaling
  - Rotation

Translations (Changing the position)

* Moving an object from one point to the other point in the co-ordinate system is called translation.

* Let us consider $P(x, y)$ is an original position, $P'(x', y')$ is a translated position

$P(x, y)$ - Original Position

$P'(x', y')$ - Translated Position

* There exist two translation parameter t_x, t_y

t_x - Translation parameter w.r.t x axis

t_y - Translation parameter w.r.t y axis

$$\boxed{\begin{aligned} x' &= x + t_x \\ y' &= y + t_y \end{aligned}} \Rightarrow \begin{aligned} x' &= x + \Delta x \\ y' &= y + \Delta y \end{aligned}$$

* Let us represent the above equation in matrix form.

$$[x' \quad y'] = [x \quad y] + [t_x \quad t_y]$$

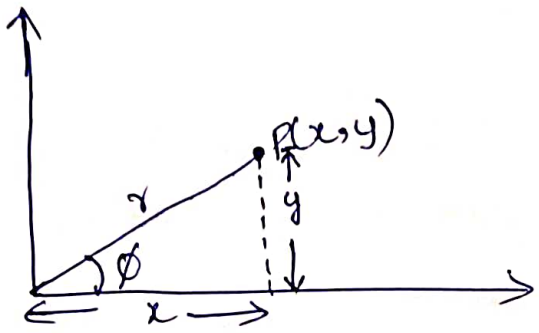
(or)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

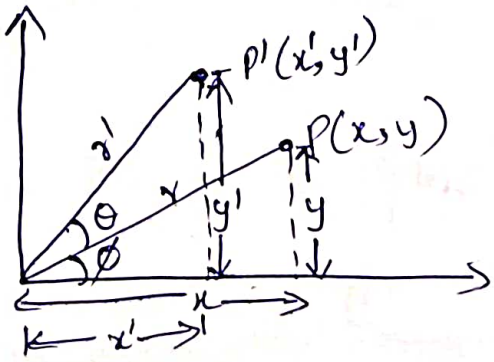
Rotation:

* Changing the angle of an object is called rotation.

* Let us assume $P(x, y)$ exist at an angle ϕ initially;



* Now, the object, is shifted its position to an angle θ



$$\cos\phi = \frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{x}{r}$$

$$\boxed{x = r\cos\phi} \quad \text{---} \quad (1)$$

$$\sin\phi = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{y}{r}$$

$$\boxed{y = r\sin\phi} \quad \text{---} \quad (2)$$

New Angle after rotation from P to P' = $(\phi+\theta)$

$$\cos(\phi+\theta) = \frac{x'}{r}$$

$$\boxed{x' = r\cos(\phi+\theta)} \quad \text{—— (3)}$$

$$\sin(\phi+\theta) = \frac{y'}{r}$$

$$\boxed{y' = r\sin(\phi+\theta)} \quad \text{—— (4)}$$

$$\cos(A+B) = \cos A\cos B - \sin A\sin B \quad \text{—— (5)}$$

$$\sin(A+B) = \sin A\cos B + \cos A\sin B \quad \text{—— (6)}$$

By utilizing these formulae in above equations, they can be written as

$$x' = r\cos\phi\cos\theta - r\sin\phi\sin\theta \quad \text{—— (7)}$$

sub eq① & ② in eq⑦

$$\boxed{x' = x\cos\theta - y\sin\theta} \quad \text{—— (8)}$$

Similarly,

$$y' = r\sin\phi\cos\theta + r\cos\phi\sin\theta$$

$$\boxed{y' = y\cos\theta + x\sin\theta} \quad \text{—— (9)}$$

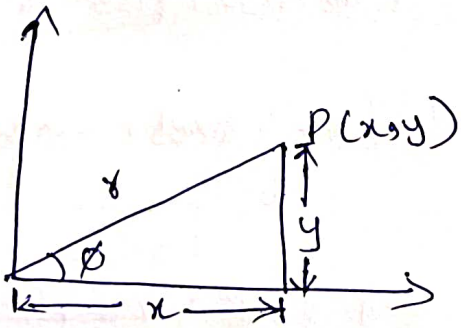
* From ϕ, the angle has been rotated to θ, the direction of rotation from ϕ to θ is anticlockwise

* Eq (8) & (9) can be represented in matrix format as shown below,

$$\begin{bmatrix} x' & y' \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \quad -(10)$$

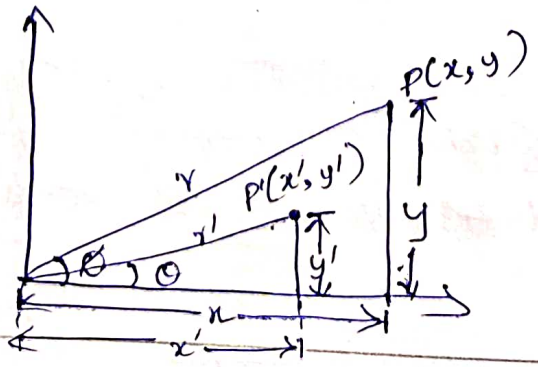
* Eq (10) represents the rotation of any point with certain angle 'θ' in anti-clockwise direction.

* Let us check the conditions for clockwise rotation



$$x = r\cos\phi$$
$$y = r\sin\phi$$

If the point is rotated to an angle 'θ' towards clockwise direction.



* New angle after rotation from P to P' = $(\phi - \theta)$

$$x' = r\cos(\phi - \theta)$$
$$y' = r\sin(\phi - \theta)$$

$\cos(A-B) = \cos A \cos B + \sin A \sin B$ } trigonometric equations
$\sin(A-B) = \sin A \cos B - \cos A \sin B$ }

↓

Modulate x' & y' by using the above equations respectively.

$$x' = r\cos\phi\cos\theta + r\sin\phi\sin\theta$$

$$\boxed{x' = x\cos\theta + y\sin\theta}$$

$$y' = r\sin\phi\cos\theta - r\cos\phi\sin\theta$$

$$\boxed{y' = y\cos\theta - x\sin\theta}$$

* Matrix Representation of x' & y'

$$\boxed{\begin{bmatrix} x' & y' \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}$$

Scaling

* Scaling is utilized to resize an object i.e., Scaling is used to either to increase (or) decrease the size of an object.

* To perform scaling two parameters S_x & S_y are required.

* If S_x & S_y is in between 0 & 1
  - Then the point is closer to origin
  - Therefore size decreases.

* If S_x & S_y is greater than 1
  - Point is away from the origin
  - Therefore size increases.

* If S_x & S_y are equal
  - Scaling is uniform (x & y axis scaled uniformly, it may either ↑ (or) ↓)

* Let us consider,

$P = (x, y)$ – Before Scaling

$P' = (x', y')$ – After Scaling

$$x' = x \cdot S_x$$
$$y' = y \cdot S_y$$

→ New Scaling Coordinates.

$$[x' \quad y'] = [x \quad y] \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$$

→ Matrix representation

$S_x \neq S_y$ – change in shape	S_x & $S_y < 1$ – size decreases
$S_x = S_y$ – No change in shape	S_x & $S_y > 1$ – size increases

Monday
07/11/2022

Examples of Basic 2D - Transformations

Translation

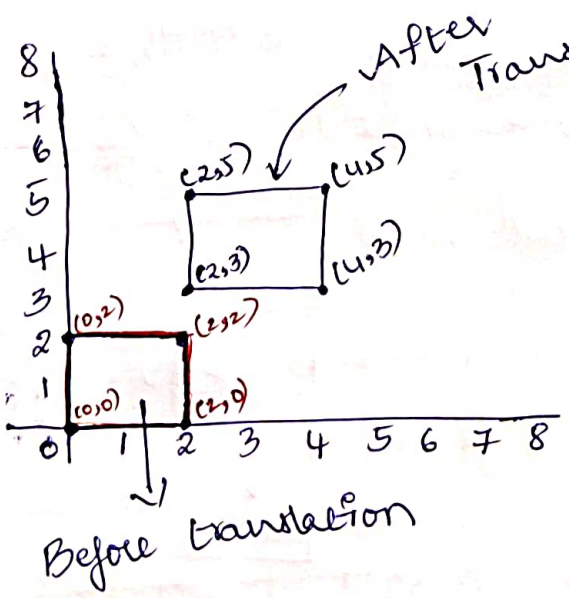
t_x, t_y, (x,y), (x',y')

$$x' = x + t_x$$
$$y' = y + t_y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

Example

Consider a square whose co-ordinates are $(0,0)$ $(2,0)$ $(0,2)$ $(2,2)$. Translation parameters $t_x=2$, $t_y=3$.



Consider $(0,0)$

$$x' = x + t_x$$
$$x' = 0 + 2 = 2$$
$$y' = 0 + 3 = 3$$

The changed coordinates are $(2,3)$

$(0,0) \longrightarrow (2,3)$

*(0,2)
x, y

$x' = 0+2 = 2$

$y' = 2+3 = 5$

$(0,2) \rightarrow (2,5)$

(2,2)
x, y

$x' = 2+2 = 4$

$y' = 2+3 = 5$

$(2,2) \rightarrow (4,5)$

(2,0)
x, y

$x' = 2+2 = 4$

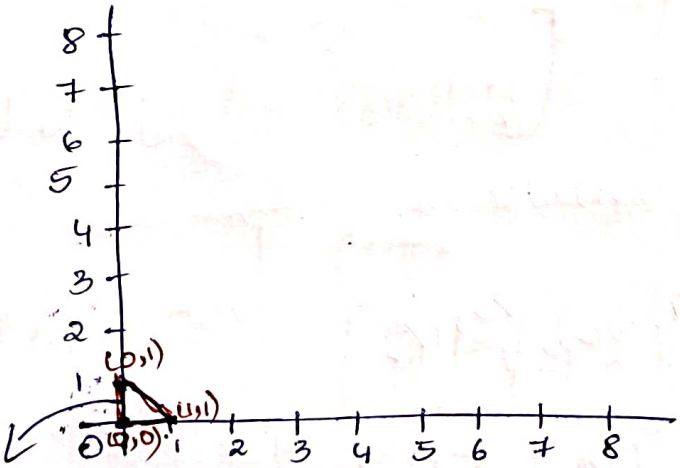
$y' = 0+3 = 3$

$(2,0) \rightarrow (4,3)$

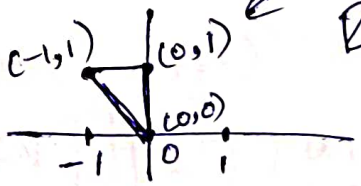
Rotation

(Anti clockwise)

Example: Let us consider a triangle whose co-ordinates are (0,0) (0,1) (1,1), $\theta = 90°$ (Anti-Clockwise Direction)



Before Rotation



After Rotation

$$[x' \; y'] = [x \; y] \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

(or)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$\sin 90° = 1$

$\cos 90° = 0$

(0,0)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$(0,0) \rightarrow (0,0)$

(1,0)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$(1,0) \rightarrow (0,1)$

(1,1)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$(1,1) \rightarrow (-1,1)$

Note:
If direction is not provided by default we need to consider it as 'anti clock wise direction'

(Clock-wise)

Example: Let us consider a triangle whose co-ordinates are (0,0) (0,1) (1,1), $\theta = 90°$ (clockwise direction)

Sol:-

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$\sin 90° = 1$
$\cos 90° = 0$

(0,0)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$(0,0) \rightarrow (0,0)$

(1,0)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

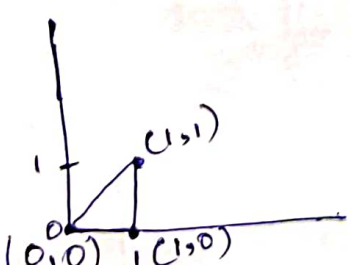
$(1,0) \rightarrow (0,-1)$

(1,1)

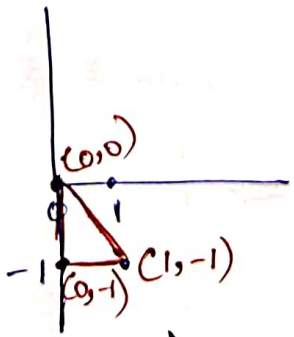
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$(1,1) \rightarrow (1,-1)$



Before Rotation



After Rotation.

Scaling

$S_x, S_y, (x,y), (x',y')$

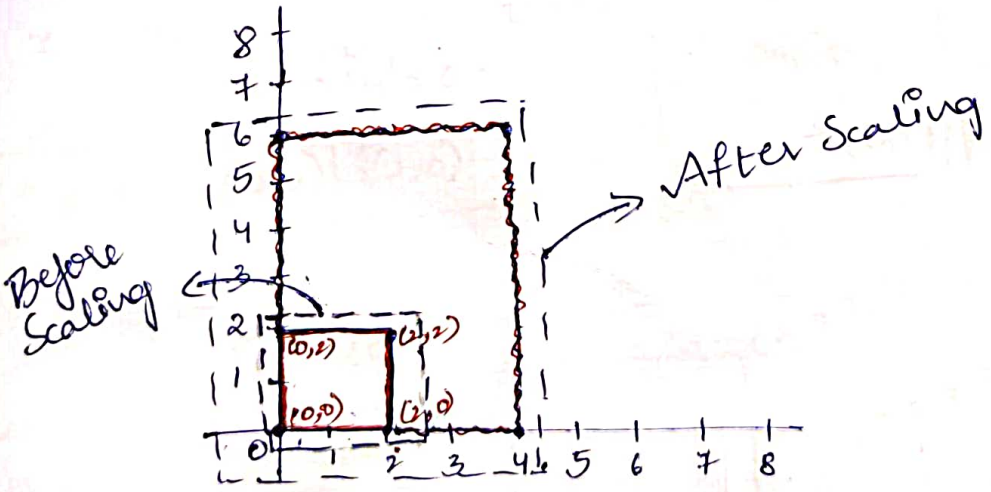
$$x' = x S_x$$
$$y' = y S_y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Example1: Consider a square whose co-ordinates are $(0,0)\ (2,0)\ (0,2)\ (2,2)$, the scaling parameter $S_x=2, S_y=3$

Sol:- * The scaling parameter are greater than 1, therefore size of the square increases

* S_x & S_y are not equal therefore the shape of the square also changes.



(0,0)
$x' = 0 \times 2 = 0$
$y' = 0 \times 3 = 0$
$(0,0) \rightarrow (0,0)$

(2,0)
$x' = 2 \times 2 = 4$
$y' = 0 \times 3 = 0$
$(2,0) \rightarrow (4,0)$

(0,2)
$x' = 0 \times 2 = 0$
$y' = 2 \times 3 = 0$
$(0,2) \rightarrow (0,6)$

(2,2)
$x' = 2 \times 2 = 4$
$y' = 2 \times 3 = 6$
$(2,2) \rightarrow (4,6)$

After Scaling

* The size is increased
* Shape is changed from square to rectangle.

Example2: Consider a square whose co-ordinates are (0,0) (2,0) (0,2) (2,2), the scaling parameter $S_x = 0.5$, $S_y = 0.5$

Sol:- * Both scaling parameters are equal, scaling is done uniformly along the x & y axis

* Shape of the object does not change.

* S_x & S_y same shape doesn't change

(0,0)

$x' = 0 \times 0.5 = 0$

$y' = 0 \times 0.5 = 0$

(0,0) → (0,0)

(2,0)

$x' = 2 \times 0.5 = 1$

$y' = 0 \times 0.5 = 0$

(2,0) → (1,0)

(0,2)

$x' = 0 \times 0.5 = 0$

$y' = 2 \times 0.5 = 1$

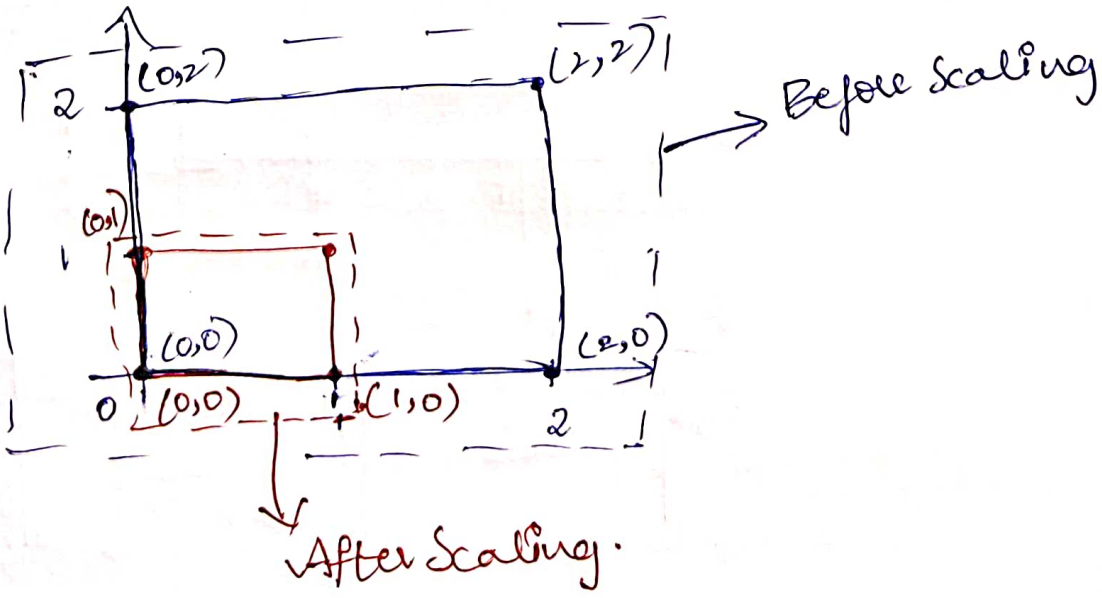
(0,2) → (0,1)

(2,2)

$x' = 2 \times 0.5 = 1$

$y' = 2 \times 0.5 = 1$

(2,2) → (1,1)



After Scaling

* Size is decreased
* Shape doesnot change